

The Unit Circle and Exact Values

ANSWER KEY



Exact values

1 Write the exact value of each expression:

a $\sin \frac{\pi}{6} = \frac{1}{2}$

b $\cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

c $\tan \frac{5\pi}{6} = -\frac{\sqrt{3}}{3}$

d $\sin \frac{3\pi}{2} = -1$

2 Write the exact value of each expression:

a $\cos \left(-\frac{\pi}{3}\right) = \frac{1}{2}$

b $\sin \frac{7\pi}{4} = -\frac{\sqrt{2}}{2}$

c $\tan \pi = 0$

d $\cos \frac{11\pi}{6} = \frac{\sqrt{3}}{2}$

Points on the unit circle

3 The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find:

a $\sin \theta = \frac{4}{5}$

b $\cos \theta = -\frac{3}{5}$

c $\tan \theta = -\frac{4}{3}$

d the quadrant containing θ Quadrant II

4 State the reference angle for each angle:

a $\frac{5\pi}{6} = \frac{\pi}{6}$

b $\frac{4\pi}{3} = \frac{\pi}{3}$

c $-\frac{\pi}{4} = \frac{\pi}{4}$

d $\frac{7\pi}{2} = \frac{\pi}{2}$

5 Identify the quadrant(s) in which each pair of conditions holds:

a $\sin \theta < 0$ and $\cos \theta > 0$ Quadrant IV.

b $\tan \theta > 0$ and $\sin \theta < 0$ Quadrant III.

6 Given that $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin \theta$ and $\tan \theta$ exactly.
 $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$, and in QIII $\sin \theta < 0$: $\sin \theta = -\frac{12}{13}$. $\tan \theta = \frac{-12/13}{-5/13} = \frac{12}{5}$.

Even/odd properties and periodicity

7 Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin \theta = 0$

b $\cos(-\theta) - \cos \theta = 0$

c $\frac{\sin(-\theta)}{\cos(-\theta)} = -\tan \theta$

8 Use periodicity to find the exact value of each expression:

a $\sin \frac{13\pi}{6} = \sin \frac{13\pi}{6} - 2\pi = \sin \frac{\pi}{6} = \frac{1}{2}$

b $\cos \frac{17\pi}{4} = \cos \frac{17\pi}{4} - 4\pi = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$

c $\tan \frac{9\pi}{4} = \tan \frac{9\pi}{4} - 2\pi = \tan \frac{\pi}{4} = 1$

Solving basic trigonometric equations on the unit circle

- 9** Find all solutions to $\sin \theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.

Reference angle $\frac{\pi}{6}$, sine negative in QIII and QIV: $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$.

- 10** Find all solutions to $2\cos \theta + \sqrt{3} = 0$ in the interval $[0, 2\pi)$.

$\cos \theta = -\frac{\sqrt{3}}{2}$: reference angle $\frac{\pi}{6}$, cosine negative in QII and QIII: $\theta = \frac{5\pi}{6}, \frac{7\pi}{6}$.